



The Bombay Salesian Society's
Don Bosco Institute of Technology
(An Autonomous Institute Affiliated to University of Mumbai)

Department of Information Technology Engineering"

Internal Assessment Test(IAT) -1 & Mid Semester Examination(MSE) from 31 Aug-04 Sep, 2026 (Odd 2026-2027)
(Maximum Marks: 30, Duration : 90 Minutes)

Syllabus Details of TE IT - SEM V Subjects

Course Code	Course Name	Faculty In-charge	Syllabus Content
25IT5PCC01	Software Engineering	Prof. Tayyabali Sayyad	1. Introduction to Software Engineering 1.1 Nature of Software, Software Engineering, Software Process, Capability Maturity Model (CMM), 1.2 Generic Process Model, Prescriptive Process Models: Waterfall Model, V-Model, Incremental Process Models, Evolutionary Process Models, Concurrent Models, 1.3 Agile Process, Agility Principles, Extreme Programming (XP), Scrum, Kanban Model 2. Requirement Analysis 2.1 Software Requirements: Functional & Non-Functional Requirements, User-System Requirement Engineering Process, Feasibility Studies, 2.2 Requirements Elicitation, Validation & Management, Software Prototyping, S/W Documentation, 2.3 Analysis and Modelling, Software Requirement Specification (SRS), Self Study Topics: Prioritizing requirements using Kano Diagram – real life application case study 3. Software Estimation and Scheduling 3.1 Management Spectrum, 3Ps (People, Product, Process), Process and Project Metrics, 3.2 Software Project Estimation: LOC, Function Points (FP), Empirical Estimation Models, COCOMO II Model, Specialized Estimation Techniques, Object-Based Estimation, Use-Case Based Estimation, 3.3 Project Scheduling: Defining a Task Set for the Software Project, Timeline Charts, Tracking the Schedule, Earned Value Analysis

25IT5PCC02	Web Technology	Dr. Vaishali Kavathekar	Module 1 : Web Programming Fundamentals 1.1 Web Communication Fundamentals : Working of the World Wide Web (WWW), Working of Web Browsers, Client–Server Communication, HTTP Protocol, HTTPS and TLS, Domain Name System (DNS), Uniform Resource Locator (URL), Uniform Resource Identifier (URI), Document Object Model (DOM), Introduction to XML and JSON, Introduction to RESTful APIs 1.2 Evolution of Modern Web Technologies: Evolution of the Web (Web 1.0, Web 2.0, Web 3.0 and Web X.0), Internet and World Wide Web (WWW), Website vs Web Application, Static vs Dynamic Websites, Multi-Page Applications (MPA) vs Single-Page Applications (SPA), Frontend, Backend and Full-Stack Development, Web Standards, Overview of Modern Web Development Ecosystem. 1.3 Modern Web Development Ecosystem: HTML5 and CSS3 Overview, Responsive Web Design Concepts, Introduction to JavaScript and ECMAScript (ES6+), Package Managers (NPM), Development Environment (VS Code, Node.js), Modern Frontend Frameworks (React, Angular and Vue), Enterprise Backend Frameworks (NestJS). Module 2: JavaScript & TypeScript 2.1 JavaScript Fundamentals: Variables (var, let, const), Data Types, Type Coercion, Operators, Control Structures, Functions, Closures, Hoisting, Scope Chain, IIFE 2.2 JavaScript OOP: Objects, Prototypes, Prototype Chain, ES6 Classes, Inheritance, Encapsulation, Polymorphism, DOM Manipulation, Event Handling, Event Bubbling 2.3 TypeScript: Static Typing, Type Annotations, Interfaces, Type Aliases, Enums, Generics, Decorators, Modules, Compilation Pipeline, tsconfig.json configuration Module 3: Asynchronous Web Programming 3.1 AJAX Fundamentals: Synchronous vs Asynchronous Processing, XMLHttpRequest (XHR) Object, readyState, onreadystatechange, HTTP Methods (GET, POST, PUT, DELETE), JSON and XML Data Formats 3.2 Fetch API: fetch(), Request/Response Objects, Headers, Handling JSON, CORS (Cross-Origin Resource Sharing), Preflight Requests, Error Handling with try/catch 3.3 Asynchronous Patterns: Callbacks, Callback Hell, Promises (resolve, reject, chaining, .then/.catch/.finally), async/await, Promise.all(), Promise.race(), Introduction to WebSockets and Real-time Data
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25IT5PEC01	Data Mining & Business Intelligence	Prof. Aruna Khubalkar	Module-1 Introduction to Data Mining 1.1 Introduction to Data Mining: What is Data Mining? Evolution of Data Mining, Data Mining vs Traditional Data Analysis 1.2 KDD Process: Stages of KDD – Data Cleaning, Integration, Selection, Transformation, Data Mining, Pattern Evaluation, Knowledge Presentation 1.3 Data Mining Systems and Applications: Types of Data Mining Systems, Classification of Data Mining Tasks, Applications in various domains Self-Learning Topics / Tools: Major Issues in Data Mining. Module-2 Data Exploration and Data Preprocessing 2.1 Data Exploration: Types of Attributes, Statistical Description of Data (Mean, Median, Mode, Variance, Skewness), Measuring Similarity and Dissimilarity 2.2 Data Preprocessing: Need for Preprocessing, Data Cleaning, Data Integration, Data Reduction (Attribute Subset Selection, Histograms, Clustering, Sampling, Data Cube Aggregation), Data Transformation and Data Discretization: Normalization, Binning, Histogram Analysis Module-3 Classification 3.1 Classification Fundamentals: Basic concepts and applications of classification in data mining 3.2 Classification Methods: 3.2.1 Decision Tree Induction – Attribute Selection Measures: Information Gain, Gain Ratio, Gini Index
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25IT5PEC03	Blockchain and Cryptocurrency	Prof. Kanchan Talekar	Module 1: Introduction to Blockchain 1.1 Structure of a Block, Block Header, Block Identifiers: Block Header Hash and Block Height, The Genesis Block, Linking Blocks in the Blockchain 1.2 Cryptographic Hash Functions, Structure of a Merkle Tree, Use of Merkle Tree in Blockchain. 1.3 Node Hierarchy, Structure of (Simplified Payment Verification) SPV Blocks, SPV Verification Protocol Module 2: Consensus and Mining 2.1 Decentralised Consensus, Byzantine General's Problem, Independent Verification of Transactions 2.2 Mining Nodes: Aggregating Transactions into Blocks, Constructing the Block Header, Successfully Mining the Block, Validating a New Block, Assembling and Selecting Chains of Blocks, Blockchain Forks Module 3: Introduction to Bitcoin 3.1 What is Bitcoin and the history of Bitcoin, getting the first Bitcoin, finding the current price of Bitcoin, and sending and receiving Bitcoin. 3.2 Bitcoin transaction, Transaction Life cycle, transaction Fees, Privacy in Transaction
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25HM5CYS E01	Ethical Hacking (Cybersecurity (Honor/Minor)	Prof. Prasad Padalkar	Module 1: Introduction to Ethical Hacking & Networks 1.1 Fundamentals of IP protocol stack, IP addressing and routing, Routing protocols, 1.2 Protocol vulnerabilities : TCP, IP, SMTP, ARP, TELNET, HTTP. 1.3 Steps of ethical hacking using References like OWASP Web Security Testing Guide (WSTG) , Open Source Intelligence (OSINT) 1.4 AI's impact on the Cyber Kill Chain, automated threat intelligence gathering, using Machine Learning for anomaly detection in routing protocols. Module 2: Applied Cryptography 2.1 Private-key & public-key encryption, Key Exchange Protocols : Kerberos, Diffie Hellman. 2.2 Cryptographic Hash Functions & applications, Digital Signature ,Digital Certificate 2.3 Steganography, biometric authentication: Finger print, Retina, Facial. 2.4 Lightweight cryptographic algorithms. : Ascon family of standards for medical devices. 2.5 AI-enhanced cryptanalysis, machine learning in biometric spoofing (e.g., Deepfakes), securing AI model weights through encryption. Module 3: Network Security & System Hacking 3.1 Information gathering, reconnaissance, scanning, vulnerability assessment (Open VAS, Nessus). 3.2 Password Cracking : Hashing vs. Encryption, salt and pepper. Types of attacks: Dictionary, Brute-force, Rule-based, and Rainbow Tables. Pass-the-Hash (PtH) concepts. Countermeasures: Multi-Factor Authentication (MFA), account lockout policies, and robust password matrices.
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25ET5MDM 01	IoT System Design (MDM)	Dr. Freda Carvalho	Module 1: Introduction to IoT Basics 1.1. Definition, characteristics, and basic components of an IoT system. Physical and logical design of IoT. 1.2 Difference between M2M (Machine to Machine) and IoT. Introduction to Cloud computing, Fog computing, and Edge computing in IoT. Module 2: IoT Hardware and Prototyping (Arduino & Raspberry Pi) 2.1 Introduction to popular IoT boards: Arduino Uno, NodeMCU (ESP8266/ESP32), and Raspberry Pi pin outs and features. 2.2 Interfacing Sensors: DHT11 (Temperature & Humidity), PIR (Motion Sensor), Ultrasonic (Distance), and MQ-2 (Gas/Smoke Sensor). 2.3 Interfacing Actuators: Relays (for switching AC home appliances), LEDs, Buzzers, and Servo Motors. 2.4 Programming Basics: Writing, compiling, and uploading basic C/C++ code using the Arduino IDE. Module 3: Wireless Technologies and IoT Protocols 3.1 Overview of IoT connectivity: Wi-Fi, Bluetooth/BLE, ZigBee, and LoRaWAN (Basic features, range, and use cases). 3.2 Introduction to IoT Application Protocols: MQTT (Publish/Subscribe, Broker, Client concept) and CoAP (Request/Response). 3.3 Comparison between HTTP and MQTT protocols for small devices.
25ET5MDM 02	Data Acquisition and Industrial Monitoring (MDM)	Dr. Madhavi Pednekar	Module 1: Sensors and Measurement Systems (1.1, 1.2,1.3 and 1.4), Module 2: Signal Acquisition and Conditioning - 2.1 Fundamentals of analog and digital signals, 2.2 Instrumentation Amplifier, 2.3 Wheatstone Bridge, 2.4 Active and Passive Filters, Module 3:Sensor Data Acquisition System - 3.1 Principles of Sampling and Data Acquisition, 3.2 Sample-and-Hold Circuits, 3.2 Data Converters, 3.3 Data Loggers and DAQ Systems,

25ME5MDM 01	Quality Engineering (MDM)	Prof. Mahesh Rajawade	1.1 Evolution and importance of quality. Dimensions of quality in products and services. Contributions of Deming, Juran, Crosby, Ishikawa, and Taguchi. Principles of Total Quality Management. 1.2 Customer focus and customer satisfaction. Employee involvement and teamwork. Continuous improvement philosophy. 1.3 Cost of Quality. 2.1 Process variation and quality characteristics. Statistical concepts for quality control. 2.2 Control Charts: $\bar{X}$ Chart, R Chart, p Chart, np Chart, c Chart, u Chart Process Capability Analysis. Process Capability Indices (C_p and C_{pk}). 3.1 Fundamentals of acceptance sampling. Single sampling plans. Producer's Risk and Consumer's Risk. Operating Characteristic (OC) Curve.
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Dr. Sunantha Guruswamy
Head of Department

Ms. Pratibha Dumane
Dean Academics